

Sam Williams

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Education

M.S. Computer Science, Virginia Tech

January 2023 – December 2023

- Thesis: *Exploring the Usability of Non-verbal Vocal Interaction (NVVI) and a Pitch-based Implementation*

B.S. Computer Science, Virginia Tech

Fall 2019 – Fall 2022

Research & Publications, Virginia Tech

Spring 2020 – Spring 2024

- Undergraduate and Graduate research at the Distributed Virtual Environments (DVE Lab), Virginia Tech
- A Relative Pitch Based Approach to Non-verbal Vocal Interaction as a Continuous and One-Dimensional Controller (HCII 2024)
- An Approach to Pitch Based Implementation of Non-verbal Vocal Interaction (IEEE VR 2024)
- Immersive Technology in the Public School Classroom: When a Class Meets (ILRN 2021)

Work Experience

Senior Member of Technical Staff, Oracle

January 2025 – Present

- Developed, maintained, and deployed UI for six high-traffic Oracle Cloud Infrastructure (OCI) plugins
 - Implemented front-end features by integrating with service-provided API specifications and contracts using Node.js, Visual Studio Code, TypeScript, React, web technologies, and playwright end-to-end testing
 - Managed the end-to-end lifecycle of features, bugs, and library updates in collaboration with service teams, product managers, and UI component library teams using Jira, Bitbucket, Confluence, Zoom and Slack
 - Presented new UX features and patterns to service teams, Product Managers (PMs), and UX review boards
 - Co-created a new multi-edit UX pattern for bulk data management now used in the OCI Full Stack Disaster Recovery service
- Architected and implemented Node.js based MCP server deployed and used by multiple teams in the Database organization
 - Automatic MCP tool generation for all RESTful HTTP operations from services' (eg, Jira) OpenAPI specification files
 - Can support thousands of MCP tools by leveraging a discovery-based tool invocation pattern
 - Allows users to extend service API clients, create custom tools, include custom headers, and other custom configurations
- Designed and developed chrome extension for debugging and testing UI/UX features
 - Leverages declarativeNetRequest (DNR) and runtime script injection to provide capabilities for request redirection, mocking data, payload/header modifications, and other network traffic manipulations
 - Co-implemented with Codex AI agent by leveraging agentic skills, an end-to-end test regression-based implementation approach, and guardrails to ensure project and functional integrity

Instructor of Record, Virginia Tech

January 2024 – May 2024

- Taught an introductory elective course on Graphical User Interfaces, including:
 - GUI Programming – concepts and technologies like MVC, event-based programming, modularity, React, MUI, etc.
 - Computer Graphics – frameworks and libraries like coordinate systems, transformations, SVG, WebGL, D3, etc.
 - Full-stack Development – MERN stack web application development (MongoDB, Express, React, Node, REST API)
- Developed classwork material and delivered lectures in a fast-paced environment to a class of 70 students
- Supervised TAs – organized weekly meetings, instructed office hours and grading procedures

Web/Asset Developer, StemPlus

Part-time, May 2020 – December 2022

- Maintained company website as a front-end developer with WordPress and HTML/JS/CSS and customized plugins
- Created 3D assets for virtual reality-based social and educational experiences in Blender and Mozilla Hubs
- Produced proprietary video content, e.g., programming tutorial videos, e-learning content, production logos

Software Engineer, ARIES Program, Virginia Tech Libraries

May 2021 – August 2021

- Developed a cross-platform web-application for desktop and Virtual Reality devices (e.g., Meta Quest)
- Collaborated with UX designers to translate user interface mockup from Figma to JavaScript GUI with Three.js

Projects

- Audio Plugin Sandbox – Quickly and interactively prototype audio plugins in a web-application sandbox environment
 - Code C/C++ based audio processing programs (signal processing) translated into JavaScript Web Audio API
 - Drag & Drop GUI interface for routing generators and effects, with input components for real-time parameters (eg, sliders)
- X86 ASM Sandbox/Debugger – Debug x86 assembly programs with breakpoints and stack and memory visualizations
 - Terminal and GUI provide ability to add breakpoints, step in instructions, or run entire program
 - Stack, registers, flags, and RAM are all visible, interactable and modifiable
- VST Production – Audio plugin development, e.g., pitch and timbre shifting, filtering and compression
 - Implemented with C++ SDK in Visual Studio, observing Model-View-Controller (MVC) principles

Other Skills

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| Programming Languages | • Typescript, JavaScript, HTML, C, C++, C#, Python, SQL, Java, Rust, x86 ASM |
| Concepts & Technologies | • UI, User Experience (UX), Audio/Digital Signal Processing, Music, Music Production, IDEs, Agentic AI, Testing, APIs, REST, Unity3D, Virtual/Extended Reality |